

A Comparative Study of Data Sampling and Generative Models for Handling Class Imbalance in Intrusion Detection Systems

Vo Thanh Nhan¹, Huynh Yen Nhi¹, Trung Ha¹, and Tran Khanh Dang^{2,3}(✉)

¹ University of Information Technology, VNU-HCM, Ho Chi Minh City, Vietnam

² Institute for Experiential Technology, Van Lang University, Ho Chi Minh City, Vietnam

³ Faculty of Information Technology, School of Technology, Van Lang University, Ho Chi Minh City, Vietnam

22521008@gm.uit.edu.vn, 22521034@gm.uit.edu.vn, trunghlh@uit.edu.vn, khanh.dt@vlu.edu.vn

(✉)The corresponding author

Abstract: This study investigates methods for addressing class imbalance in the CICIDS2017 network intrusion detection dataset. The dataset contains 78 traffic features with both benign and multiple attack categories. We applied three synthetic data generation approaches—SMOTE, CGAN, and Variational Autoencoder (VAE). SMOTE interpolates minority class samples, CGAN generates conditioned synthetic data via adversarial training, and VAE models the latent probability distribution to create realistic samples. We apply two machine learning algorithms to the augmented datasets—Random Forest (RF) and XGBoost (XGB), with training (80%) and testing (20%) subsets. Experimental results showed that SMOTE and CGAN, when combined with RF and XGB, achieved comparable classification performance, achieving accuracies of up to 99% and F1-scores above 0.95 across several minority attack types, while VAE lagged behind with lower predictive performance. These findings demonstrate the effectiveness of combining synthetic data generation with machine learning algorithms for multi-class intrusion detection in imbalanced datasets. However, this study is confined to traditional classifiers; future research will extend the evaluation to deep learning models and federated learning frameworks to achieve greater robustness and scalability in real-world IDS applications.

Keywords: Intrusion Detection · Class Imbalance · SMOTE · SMOTE · CGAN · Variational Autoencoder · Random Forest · XGBoost · CICIDS2017.

1 Introduction

With the fast growth of Internet of Things (IoT) devices and the adoption of 5G networks, intrusion detection systems (IDS) have become increasingly evident in safeguarding digital infrastructure against increasingly sophisticated cyber

threats, including Advanced Persistent Threats (APTs), zero-day attacks and ransomware. In this evolving landscape, IDS must effectively identify abnormal activities amidst massive volumes of legitimate traffic. However, one major challenge in addressing the intrusion detection problem lies in the fact that attack samples often account for only a small portion of overall network traffic, leading to severe data imbalance. As a result, machine learning models frequently exhibit bias toward the majority classes, impairing their ability to identify uncommon but dangerous behaviors.

Over the past decade, the utilization of machine learning (ML) in IDS has grown considerably owing to its capability to capture complex attack patterns in network data. Recent works have employed supervised learning, unsupervised clustering, and deep learning architectures to enhance intrusion detection accuracy. A study evaluates XGBoost and Random Forest classifiers combined with three upsampling techniques (SMOTE, ADASYN, GNUS) for churn prediction under varying class imbalance levels, revealing that tuned XGBoost with SMOTE consistently performs best, while Random Forest struggles under severe imbalance [1]. Another study proposes LCCDE, an ensemble IDS framework that selects the best-performing model per attack type using XGBoost, LightGBM, and CatBoost, leveraging prediction confidence for accurate intrusion detection in IoV networks, and demonstrates its effectiveness on Car-Hacking and CICIDS2017 datasets [2]. Despite these advances, class imbalance—where minority attack types are greatly outnumbered by normal traffic—remains a critical limitation that hinders real-world applicability. Specifically, CICIDS2017 – a dataset contains a total of 78 detailed features describing network traffic, including both normal traffic (BENIGN) and six types of attacks: DoS, Web Attack, Bot, PortScan, Brute Force, and Infiltration. Among these types, Brute Force and Infiltration are significantly underrepresented but high-risk attack classes, leading to model bias toward the majority class and poor generalization to minority intrusions.

This imbalance can result in elevated false negative rates, posing severe risks in operational environments where undetected attacks can escalate rapidly. Although several traditional oversampling methods have been proposed—such as SMOTE and ADASYN—these often fail to fully capture the complex distributions of minority classes. To address this, generative models such as Conditional GAN (CGAN) and Variational Autoencoder (VAE) have been explored in recent studies for their ability to perform conditional synthetic data generation, approximating real data distributions more effectively. However, there remains a lack of comprehensive evaluations that compare the performance of these generative augmentation methods against traditional techniques like SMOTE within the context of multi-class imbalance in IDS datasets.

The research question posed in this study is: Can generative models such as CGAN and VAE enhance the performance of network attack detection models compared to SMOTE when addressing the multi-class imbalance problem in the CICIDS2017 dataset?

Therefore, this paper aims to:

1. Evaluate the effectiveness of SMOTE, CGAN, and VAE in mitigating the issue of class imbalance within the CICIDS2017 dataset.
2. Compare the performance of XGBoost and Random Forest in multi-class classification after applying these augmentation methods.
3. Investigate the impact of different normal-to-attack traffic ratios (1:1, 1:2, 1:3, 1:4, and 1:5) on detection accuracy and generalization.

To address the research objectives in a structured manner, the study is organized as follows. A survey of related work is provided in Section 2. Section 3 describes the dataset used in this study, details the synthetic data generation methods (SMOTE, CGAN, and VAE), and introduces the machine learning classifiers (XGBoost and Random Forest) applied for evaluation. Section 4 presents the experimental results and evaluates the models’ performance. Finally, Section 5 summarizes the work and outlines directions for future research.

2 Literature Review

Machine learning enhances Intrusion Detection Systems (IDS) by learning patterns from network traffic to detect both known and novel attacks. Nonetheless, intrusion detection datasets frequently exhibit pronounced class imbalance, with minority attack types being insufficiently represented. Such imbalance tends to skew models toward the dominant classes, thereby weakening their capability to recognize critical threats and driving the adoption of resampling or generative methods to enhance detection of underrepresented classes. [3]. However, in [4], the authors noted several challenges in applying generative models to imbalanced data, such as difficulties in capturing complex minority attack patterns, risks of generating unrealistic or redundant samples, and vulnerability to mode collapse. Training stability and sensitivity to hyperparameters further complicate their use, while the trade-off between realism and diversity often limits improvements in IDS performance. Ethical concerns regarding potential misuse of synthetic data also remain.

Building on this observation, in [5], the authors analyzed how the imbalance rate (IR) affects the effectiveness of machine learning models, applying three resampling strategies—oversampling, undersampling, and hybrid sampling—across 48 binary imbalanced datasets, including CICIDS2017. The IR was progressively reduced while maintaining the original data distribution, and the results showed that classification performance generally improved as IR decreased. Models such as Random Forest demonstrated higher stability compared to other algorithms. However, this study provided only an overall evaluation for each dataset without a detailed analysis of the influence of individual IR levels on data quality, nor did it investigate generative augmentation methods beyond SMOTE.

Similarly, in [6], the authors introduced an oversampling technique leveraging a Conditional Generative Adversarial Network (CGAN) to address class imbalance in the NSL-KDD and UNSW-NB15 datasets. The method generated synthetic minority class samples conditioned on class labels, aiming to increase

the effectiveness in detecting rare attacks. The impact of data balancing was assessed using eight machine learning classifiers, with t-SNE visualization employed to assess the quality of the synthetic samples. Results demonstrated that CGAN-generated data significantly improved minority class detection, leading to an increase of up to 8% in prediction accuracy and approximately 13% in the Matthews Correlation Coefficient (MCC) compared with models relying on the imbalanced original data.

In another study, [7] investigated how two oversampling methods, SMOTE and ADASYN, influence the classification performance of four algorithms—Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors—on the UNSW-NB15 dataset. The study also examined the effect of applying data balancing either before or after feature selection on model accuracy. Results indicated that SMOTE produced higher-quality synthetic samples for this dataset compared to ADASYN. Furthermore, applying Recursive Feature Elimination (RFE) improved training efficiency but caused a slight, negligible decrease in accuracy. The study was limited to a single dataset, potentially limiting generalization to datasets with higher imbalance ratios, and only considered simple oversampling methods without exploring more advanced generative models.

Similarly, [8] addressed credit card fraud detection, where fraudulent transactions are severely underrepresented. Using federated learning, banks collaboratively trained a global model without sharing sensitive data, obtaining an accuracy as high as 97% when evaluated on the European Credit Card dataset. This highlights the potential of collaborative and privacy-preserving approaches to tackle imbalance and security issues, similar to those faced in IDS.

Furthermore, in [9], the authors evaluated machine learning algorithms under various imbalance ratios (1:10, 1:100, 1:500, and 1:1000) using different resampling strategies on the CIC-DDoS2019 dataset. The study revealed that Under-Over Bagging and SMOTEBoost effectively maintained equilibrium between precision-recall and sensitivity-specificity across multiple levels of class imbalance. In the case of a 1:10 imbalance ratio, Random Forest achieved an F1-score of 0.9996, but performance declined as the imbalance ratio increased, particularly at 1:500 and 1:1000. This degradation was attributed to random undersampling, which can lead to overfitting on minority classes or inadequate learning of majority classes. The study was limited to binary classification on a single dataset and relied on a simple generative approach (SMOTE).

In contrast, in [10], the authors proposed MAIDS, an intrusion detection system for SDN-IoT networks, employing XGBoost and Random Forest to enhance detection performance and reduce false alarms. The study achieved mean accuracy scores of 99.88% and 99.89% for XGBoost and Random Forest, respectively, and emphasized the importance of dataset selection, identifying CICIDS2017 as more effective than NSL-KDD for reducing false positives and false negatives. However, the issue of class imbalance and its impact on detection performance was not addressed.

Table 1: Comparative Summary of Related Work on ML-Based IDS

Ref	Dataset(s)	Method(s)	Key Results	Limitations
[5] Zheng et al.	48 binary imbalanced datasets (including CICIDS2017)	Oversampling, undersampling, hybrid sampling	Performance generally improved as IR decreased; RF showed high stability	No detailed analysis of individual IR levels; no generative methods beyond SMOTE
[6] Dina et al.	NSL-KDD, UNSW-NB15	Conditional GAN (CGAN) oversampling	Improved minority detection; +8% accuracy, +13% MCC vs. imbalanced	Limited to two datasets; no analysis of varying IR levels
[7] Barkah et al.	UNSW-NB15	SMOTE, ADASYN; RFE feature selection	SMOTE produced better samples; RFE improved training efficiency	Single dataset; simple oversampling methods only
[9] Shanmugam et al.	CIC-DDoS2019	Under-Over Bagging, SMOTEBoost	Balanced precision-recall and sensitivity-specificity; RF F1=0.9996 at IR 1:10	Binary classification only; simple SMOTE-based generation
[10] Ferrão et al.	CICIDS2017, NSL-KDD	XGBoost, Random Forest (MAIDS IDS)	High accuracy (99.88% XGB, 99.89% RF) in SDN-IoT	Did not address class imbalance
[11] Lopez-Martin et al.	NSL-KDD	Variational Autoencoder (VAE) generation	Improved accuracy and F1 for minority classes; more stable than GANs	Single dataset; no varying IR evaluation

Extending the exploration of generative approaches, [11] proposed a data generation framework built upon the Variational Autoencoder (VAE) to address the problem of data imbalance in intrusion detection systems. The NSL-KDD dataset was employed to develop the model, where VAE was used to create synthetic data to augment minority classes, aiming to improve detection performance across different attack categories. The study compared the proposed approach with traditional oversampling methods such as SMOTE and ADASYN, as well as GAN-based techniques. Experimental results demonstrated that VAE-generated data improved classification accuracy and F1-scores for minority classes, while maintaining competitive performance for majority classes. The authors also emphasized that VAE offers better training stability compared to GAN-based models. However, the study was limited to a single dataset and did not evaluate the effect of varying imbalance ratios, leaving its generalizability to other datasets and imbalance scenarios untested.

As shown in Table 1, recent studies have shown that machine learning models based on ensembles, such as Random Forest and XGBoost, offer strong and reliable performance in intrusion detection systems. Nevertheless, the challenge of data imbalance continues to hinder their overall effectiveness, particularly

in terms of generalizability and classification accuracy. Although methods like SMOTE, CGAN, and VAE have shown potential in addressing this issue, their performance is inconsistent across different datasets. This emphasizes the need for a comprehensive and systematic evaluation of generative techniques under various class distribution scenarios—an area this study aims to investigate.

3 Research Method

3.1 Overview of Research Objectives and Methodology

This research aims to tackle the class imbalance issue in Intrusion Detection Systems (IDS), a factor that often diminishes the capability of machine learning algorithms when dealing with rare or infrequent attack types. The main task is to evaluate the effectiveness of minority-class data generation techniques in enhancing the detection capability for underrepresented attack categories while maintaining high predictive performance across the entire dataset. The study follows a structured process that includes the following steps: data preprocessing, application of synthetic data generation techniques (SMOTE, CGAN, VAE), dividing the dataset into training and testing portions, models were subsequently trained using Random Forest and XGBoost, and finally evaluating performance with metrics, namely accuracy, precision, recall and f1-score.

3.2 Data Preprocessing Steps

The dataset used in this research is CICIDS2017 (UNB, 2017), consisting of 78 features and a label column (BENIGN and multiple attack types such as DoS, WebAttack, Bot, PortScan, BruteForce, Infiltration). Before model training, Random Forest is employed to compute the feature importance scores. The top 10 and top 20 most important features are then selected, and models are trained using XGBoost and Random Forest to compare performance in terms of accuracy, training time, and memory usage to identify the optimal configuration. Results indicate that using the top 10 features yields high accuracy, shorter training time, and lower resource consumption, and thus is chosen for the final training phase.

3.3 Synthetic Data Generation Techniques

SMOTE (Synthetic Minority Over-sampling Technique) [12]: A method that performs linear interpolation between minority-class samples to generate new instances, thereby increasing data representativeness without duplication.

SMOTE creates synthetic samples by applying interpolation techniques between existing minority-class samples, based on the following mathematical formula:

$$Z = X_i + (\hat{X}_i - X_i) * \delta \quad (1)$$

Where:

- Z is the newly generated data point.
- X_i represents the feature set of a minority-class sample.
- $\hat{X}_i$ denotes one of the k nearest neighbors of X_i .
- δ is a random number in the range $[0, 1]$.

In this research, SMOTE is applied by selecting the k nearest neighbors for each minority-class instance, then creating new synthetic samples via linear interpolation, aiming to balance the dataset before training.

CGAN (Conditional Generative Adversarial Network) [13]: A conditional version of GAN that enables the generation of data based on specific class labels. CGAN learns the distribution of minority-class data and produces new samples that are both realistic and class-consistent.

- **Generator:** Takes random noise combined with conditional information (e.g., class label or feature input) and generates synthetic data that aligns with the given condition.
- **Discriminator:** Receives both real data with its corresponding condition and generated data with the same condition, learning to distinguish between them while considering class consistency.

VAE (Variational Autoencoder) [14]: The core structure of a generative model consists of two components: an encoder and a decoder. In the case of a Variational Autoencoder (VAE), approximate inference is performed, and the model parameters are optimized via gradient descent. The encoder maps the input into a lower-dimensional latent space, producing a continuous representation. From this latent embedding, the decoder reconstructs the data, enabling the model to synthesize new samples that closely reflect the original distribution.

During the experiment, the training dataset was input into the VAE. The encoder calculates z_{mean} and z_{log_var} , and the reparameterization trick is applied to produce the latent embedding z . The decoder then reconstructs the data from z . To reduce the gap between the original input and the reconstructed data, the model is trained with the MSE loss function. Once training is complete, synthetic samples are generated by drawing points from the latent space and passing them through the decoder.

To visually assess the similarity in data distribution between original and synthetic data generated by SMOTE, CGAN, and VAE, we utilized the popular dimensionality reduction method, t-Distributed Stochastic Neighbor Embedding (t-SNE). [6]. The datasets (both original and generated) were combined and reduced to 2D for visualization, allowing comparison of the feature space distribution and supporting the evaluation of synthetic data quality. In addition to visualization, quantitative evaluations were performed to measure the statistical discrepancy between datasets. Specifically, pairwise comparisons among the generated datasets (SMOTE, CGAN, VAE) were performed using k-Means,

silhouette score, and average entropy. These metrics allow for a rigorous assessment of how closely the synthetic samples approximate each other in terms of distributional characteristics. The combined use of both visual and statistical evaluations strengthens the reliability of the analysis and ensures a comprehensive comparison across different data generation methods.

For the CGAN and VAE models, a grid search strategy was applied to optimize hyperparameters and improve the fidelity of the generated data. Multiple combinations of key hyperparameters, including latent dimension, learning rate, batch size and training epochs, were systematically explored. Each model was trained under these settings and evaluated using Mean Squared Error (MSE), Wasserstein Distance (WD), and Kullback–Leibler (KL) Divergence. The parameter set achieving the best overall score was then used to generate the final synthetic dataset. This procedure ensures that the synthetic data from CGAN and VAE closely approximates the distribution of the original data, enhancing the reliability of both quantitative and visual comparisons.

3.4 Machine Learning Models for Evaluation

XGBoost (Extreme Gradient Boosting) [15]: An advanced boosting algorithm with high predictive power, regularization to reduce overfitting, and ability to handle sparse data efficiently. It supports customizable objective and evaluation functions, parallel computation, and deployment across various platforms and programming languages. Due to its scalability, speed, and flexibility, XGBoost has been widely adopted in real-world applications, making it a strong choice for building robust intrusion detection systems in modern network environments.

Random Forest [16]: A learning algorithm that creates many decision trees using random subsets of data and merges their predictions for the final outcome. This model can handle imbalanced data, missing values, and reduce the risk of overfitting, thereby improving generalization capability. Random Forest is well known for its efficiency and accuracy, and is widely applied in classification, data prediction, and supervised learning.

Both models are trained on data balanced using the synthetic generation techniques (SMOTE, CGAN, VAE) at different ratios (1:1, 1:2, 1:3, 1:4, 1:5), and assessed based on accuracy, precision, recall, f1-score to determine the optimal configuration.

4 Empirical Evaluation

4.1 Experimental Setup

The study’s experiments were run on the Google Colab platform within a Python 3.9 virtual environment. “All computations were carried out on the CPU, with

the implementation leveraging Python libraries including pandas, numpy, matplotlib, and seaborn, scikit-learn, XGBoost, imbalanced-learn, TensorFlow/Keras, and PyTorch, which supported tasks such as data preprocessing, synthetic data generation (SMOTE, CGAN, VAE), model training (Random Forest, XGBoost), and result visualization.

Table 2: Clustering quality evaluation of data augmentation methods (SMOTE, CGAN, VAE) under different imbalance ratios.

Ratio	SMOTE			CGAN			VAE		
	Inertia	Silhouette	Best K	Inertia	Silhouette	Best K	Inertia	Silhouette	Best K
1:1	480.52	0.7516	7	505.19	0.7226	10	1383.60	0.5763	4
1:2	611.69	0.7674	8	555.38	0.7956	10	1624.08	0.4371	6
1:3	1525.73	0.7662	4	530.99	0.8175	10	3365.24	0.3971	3
1:4	1765.63	0.7749	5	1566.61	0.8556	6	4045.38	0.3642	3
1:5	2213.84	0.7821	4	477.24	0.8941	10	4755.68	0.3499	3

For synthetic data generation, SMOTE was applied using its default configuration, which balances the minority class without altering the neighborhood structure or distance metric ($k = 5$, `random_state = 42`). The optimal CGAN setup utilized a latent space of dimension 8, mini-batch sizes of 64, 300 training epochs, and was optimized using Adam with a learning rate of 0.0005. The input noise vector was drawn from a standard normal distribution, and binary cross-entropy served as the loss function for both the generator and the discriminator. The VAE model adopted a densely connected encoder-decoder model with a latent dimension of 32, trained over 100 epochs using a batch size of 32 and the Adam optimizer with a learning rate of 0.001. Its loss function combined mean squared error for reconstruction with a KL divergence regularization term.

To thoroughly evaluate the realism of the synthetic dataset, we employed K-means clustering metrics (inertia and silhouette score), Average Entropy, and t-SNE visualization in addition to the classification metrics. These methods allow us to evaluate data compactness, diversity, and similarity to the original distribution, providing both quantitative and qualitative insights into synthetic data quality.

Regarding classification, Random Forest was configured with 100 trees using default depth and a fixed random seed (42) to ensure reproducibility. XGBoost was optimized with 100 estimators, a 0.1 learning rate, tree depth capped at 3, and log-loss for evaluation. All hyperparameters were kept consistent across experimental settings to ensure fair comparisons.

Based on a preliminary feature selection process described in Section 3.2, we assessed the effect of changing the quantity of input features. The analysis showed that selecting the top 10 most important features provided a favorable balance between predictive performance and model complexity. Therefore, this reduced feature set was used in all subsequent experiments.

Table 3: Comparison of average entropy values across SMOTE, CGAN, and VAE under different imbalance ratios.

Imbalanced Ratio	SMOTE	CGAN	VAE
1:1	0.8826	0.9928	1.4541
1:2	0.8914	1.1377	1.7864
1:3	0.8885	1.0828	1.9147
1:4	0.8844	1.1850	1.9355
1:5	0.8794	0.9281	1.9599

We partitioned the dataset into training (80%) and testing (20%) subsets, using stratified sampling to retain the class distribution. As mentioned in [2], the authors generated synthetic data on the CICIDS2017 dataset with imbalance ratios ranging from 1:4 to 1:1, where the ratios were rounded down. In our work, we evaluate the same set of ratios and additionally include a 1:5 ratio, where the rounding is performed upward. Each scenario was evaluated using identical synthetic data generation methods and classifiers to ensure consistent conditions across experiments.

As outlined in Section 3.4, model performance in these scenarios was assessed using accuracy, precision, recall, and f1-score.

4.2 Evaluation of Data Augmentation Methods: SMOTE, CGAN, and VAE

To compare the clustering performance of synthetic data generated by SMOTE, CGAN, and VAE under different class balance ratios, we evaluated the samples using inertia and silhouette metrics, with each model assessed at its optimal k . As shown in Table 2, at the balanced ratio (1:1), SMOTE and CGAN yield substantially lower inertia values than VAE, indicating more compact clusters, with SMOTE performing best. SMOTE also achieves the highest silhouette score (0.7516), reflecting well-connected and clearly separated clusters. As imbalance increases, however, CGAN demonstrates superior performance, often attaining the highest silhouette scores that further improve with more severe imbalance, highlighting its ability to preserve cluster boundaries under challenging conditions. In contrast, VAE consistently produces higher inertia and lower silhouette scores, suggesting more scattered and less distinguishable clusters, likely due to weaker learning of the underlying distribution. Overall, SMOTE is strong in stability and compactness, while CGAN excels in generating diverse yet well-separated clusters as imbalance grows.

Average Entropy is a metric that reflects the diversity of data distribution within clusters, with higher values indicating that clusters contain more heterogeneous data points, whereas excessively low values typically suggest more homogeneous clusters. The Average Entropy values for each model across different imbalance ratios are summarized in Table 3. It is apparent that SMOTE consistently exhibits low and stable Average Entropy values across all imbalance ratios. In contrast, VAE consistently shows the highest values, indicating that

its clusters are less homogeneous compared to those produced by SMOTE and CGAN.

We used t-SNE to illustrate both real and synthetic data generated by SMOTE, CGAN, and VAE across different imbalance ratios (Figs. 1–5). SMOTE consistently produced distributions closest to the original data, preserving cluster boundaries and structures with limited added diversity. CGAN generated more dispersed points and denser clusters, expanding feature space and enhancing diversity while maintaining overall class separation. VAE, in contrast, produced smoother and more interleaved distributions, often reshaping clusters and introducing overlaps; this increased diversity but reduced cohesion. As the imbalance ratio became more severe, SMOTE emphasized stability, CGAN showed stronger structural shifts with greater diversity, and VAE filled sparse regions while softening class boundaries. Class labels are represented as: 0–Benign, 1–Bot, 2–Brute-Force, 3–DoS, 4–Infiltration, 5–PortScan, 6–WebAttack.

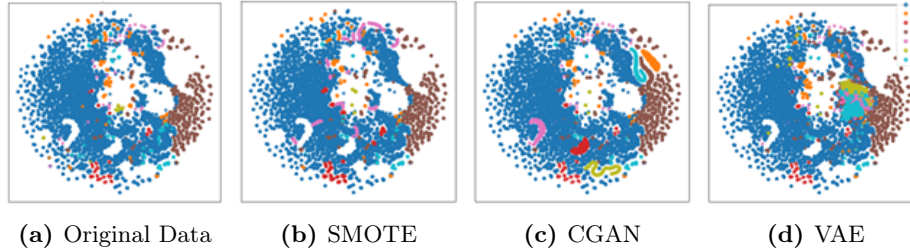


Fig. 1: t-SNE Visualization of Generated Data at 1:1 Balanced Ratio.

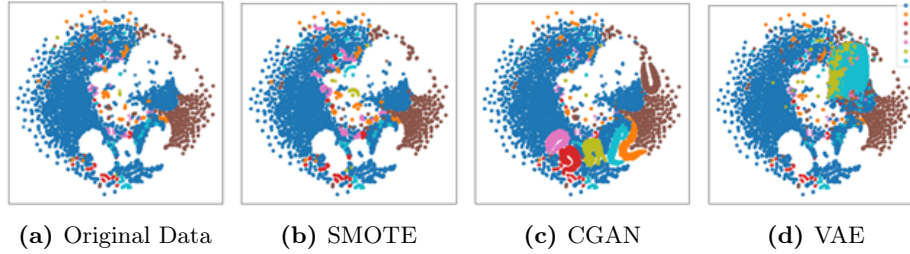


Fig. 2: t-SNE Visualization of Generated Data at 1:2 Imbalanced Ratio.

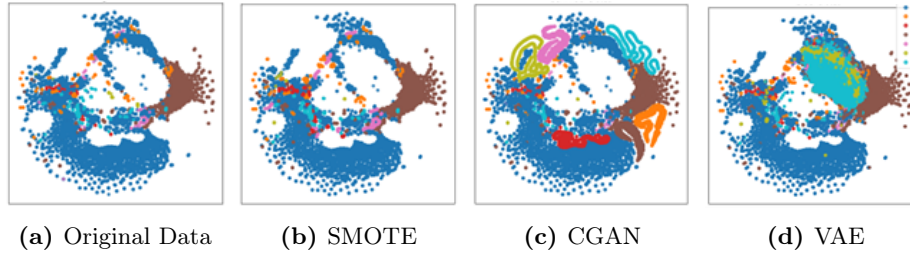


Fig. 3: t-SNE Visualization of Generated Data at 1:3 Imbalanced Ratio.

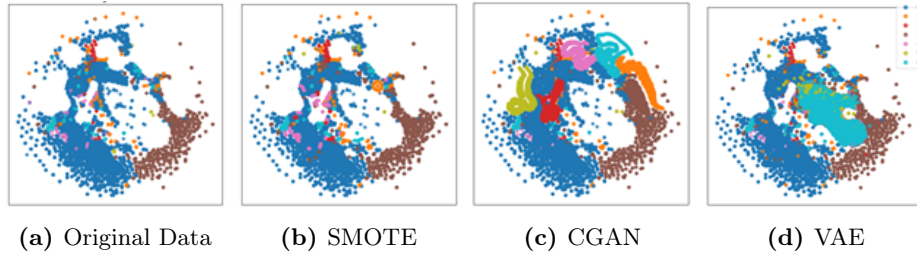


Fig. 4: t-SNE Visualization of Generated Data at 1:4 Imbalanced Ratio.

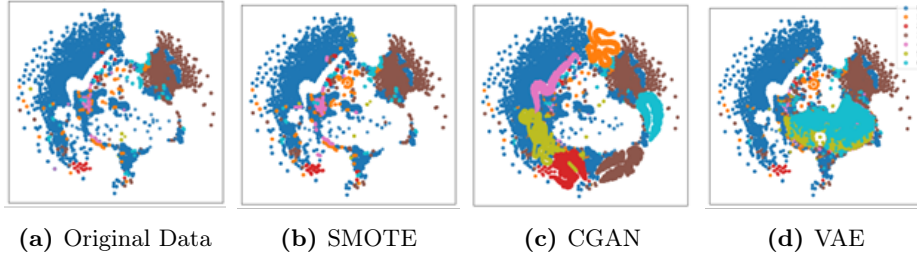


Fig. 5: t-SNE Visualization of Generated Data at 1:5 Imbalanced Ratio.

4.3 Evaluating Class Balance Ratios and Multi-Class Classification Performance of XGBoost and Random Forest

Table 4: Comparison of Accuracy between XGBoost and Random Forest (RF) across Imbalanced Ratios (SMOTE, CGAN, VAE).

Ratio	SMOTE		CGAN		VAE	
	RF	XGBoost	RF	XGBoost	RF	XGBoost
1:1	0.9907	0.9907	0.9921	0.9933	0.9155	0.9186
1:2	0.9929	0.9921	0.9946	0.9946	0.8334	0.8364
1:3	0.9955	0.9943	0.9967	0.9958	0.7467	0.7490
1:4	0.9962	0.9958	0.9970	0.9963	0.7296	0.7364
1:5	0.9967	0.9960	0.9975	0.9969	0.7126	0.7246

To obtain further insights into how class distribution influences model performance, we evaluate different class balance ratios and compare the multi-class classification effectiveness of XGBoost and Random Forest using not only Accuracy but also Precision, Recall, and F1-score. The summarized results are presented in Table 4 (Accuracy), Table 5 (Precision), Table 6 (F1-score), and Table 8 (Recall).

Overall, the experimental results across all four metrics consistently demonstrate the effectiveness of SMOTE and CGAN in supporting both Random Forest and XGBoost, whereas VAE exhibits considerably weaker performance. Specifically, SMOTE enables both classifiers to maintain highly stable results regardless of the imbalance ratio, with Random Forest showing a slight advantage under

more imbalanced conditions. CGAN achieves the best overall performance, with XGBoost performing slightly better on balanced or near-balanced data (e.g., 1:1 and 1:2), while Random Forest surpasses XGBoost as the imbalance becomes more severe (e.g., 1:3 to 1:5).

Table 5: Comparison of Precision between XGBoost and Random Forest (RF) across Imbalanced Ratios (SMOTE, CGAN, VAE).

Ratio	SMOTE		CGAN		VAE	
	RF	XGBoost	RF	XGBoost	RF	XGBoost
1:1	0.9879	0.9874	0.9906	0.9920	0.8118	0.8158
1:2	0.9922	0.9907	0.9949	0.9952	0.7690	0.7708
1:3	0.9945	0.9909	0.9967	0.9961	0.7041	0.7040
1:4	0.9947	0.9938	0.9971	0.9967	0.7115	0.7170
1:5	0.9953	0.9938	0.9976	0.9970	0.7021	0.7145

In contrast, VAE consistently yields the lowest scores across all metrics, with a noticeable decline as the imbalance ratio increases, indicating its limited ability to generate high-quality synthetic data. Taken together, these findings suggest that while both SMOTE and CGAN are effective in mitigating class imbalance, CGAN delivers superior performance across metrics, and Random Forest demonstrates greater robustness than XGBoost under heavily imbalanced scenarios.

Table 6: Comparison of F1-score between XGBoost and Random Forest (RF) across Imbalanced Ratios (SMOTE, CGAN, VAE).

Ratio	SMOTE		CGAN		VAE	
	RF	XGBoost	RF	XGBoost	RF	XGBoost
1:1	0.9906	0.9906	0.9927	0.9937	0.8140	0.8184
1:2	0.9927	0.9921	0.9955	0.9955	0.7687	0.7717
1:3	0.9952	0.9932	0.9973	0.9965	0.7008	0.7024
1:4	0.9957	0.9948	0.9974	0.9968	0.7065	0.7137
1:5	0.9953	0.9935	0.9977	0.9971	0.6989	0.7118

Table 7: Comparison of training time between XGBoost and Random Forest (RF) across Imbalanced Ratios (SMOTE, CGAN, VAE).

Ratio	SMOTE		CGAN		VAE	
	RF	XGBoost	RF	XGBoost	RF	XGBoost
1:1	1.9519s	1.4340s	2.1385s	1.4191s	4.4836s	2.1267s
1:2	2.3855s	2.5748s	3.9538s	2.4497s	10.6973s	6.4186s
1:3	3.3722s	3.1376s	8.4718s	3.4269s	21.2168s	6.6750s
1:4	5.0487s	4.9424s	11.2371s	4.1393s	32.2190s	6.1688s
1:5	6.2894s	7.0123s	11.9142s	7.2129s	37.1446s	9.1858s

We analyzed the training time of Random Forest (RF) and XGBoost (XGB) using datasets generated by SMOTE, CGAN, and VAE under different imbalance ratios (Table 7). In general, higher imbalance led to longer training times due to increased data volume. CGAN produced the fastest training for XGB at

ratios 1:1 and 1:4, while SMOTE was faster for RF at 1:2 and 1:5, and for XGB at 1:3. RF was more sensitive to the data generation method, particularly with VAE, which consistently required longer training. In contrast, XGBoost showed greater stability, with training times less affected by the type of synthetic data.

Table 8: Comparison of Recall between XGBoost and Random Forest (RF) across Imbalanced Ratios (SMOTE, CGAN, VAE).

Ratio	SMOTE		CGAN		VAE	
	RF	XGBoost	RF	XGBoost	RF	XGBoost
1:1	0.9933	0.9938	0.9948	0.9954	0.8168	0.8215
1:2	0.9932	0.9935	0.9961	0.9958	0.7720	0.7744
1:3	0.9959	0.9956	0.9979	0.9970	0.7026	0.7038
1:4	0.9966	0.9957	0.9976	0.9969	0.7066	0.7134
1:5	0.9954	0.9933	0.9978	0.9972	0.7003	0.7124

5 Conclusion

Reliable intrusion detection requires balanced and representative data, yet in practice IDS training sets are often dominated by benign traffic while rare but critical attacks are severely underrepresented. This imbalance undermines the ability of IDS to detect the very threats that matter most, making such systems less trustworthy in real-world deployment. Importantly, this challenge extends beyond IDS to many other machine learning applications where minority classes carry the greatest significance. To address this, we evaluated the effectiveness of synthetic data generated by SMOTE, CGAN, and VAE on the CICIDS2017 dataset under varying imbalance ratios. Among the evaluated methods, SMOTE showed reliable performance under mild imbalance, while CGAN achieved the best classification results in more severe scenarios. By contrast, VAE was less effective, indicating its limited ability to capture the underlying distribution. We also found that Random Forest consistently outperforms XGBoost, indicating its robustness for IDS deployment under imbalanced conditions. These findings suggest that generative approaches, especially CGAN, hold promise for enhancing the detection of rare attacks, while tree-based ensembles like Random Forest remain a practical choice in real-world systems. Future research should extend these insights to deep learning and federated settings, and explore applications beyond IDS where reliable detection of minority classes is equally critical.

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